MILITARY SPECIFICATION

TERMINAL, TELEGRAPH, AN/UGC-1

1. SCOPE

1.1 This specification covers Telegraph Terminal Set AN/UGC-1, comprising an electronic transmitting and receiving, time-division, multiplex equipment for use on radio circuits or wire lines.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.
MIL-P-17555 - Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-189 - Racks, Electrical Equipment, 19-Inch and Associated Panels.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - The material used in constructing the equipments shall be as specified in Specification MIL-E-16400. Particular attention should be paid to the acceptable and unacceptable material requirements of Specification MIL-E-16400.

3.1.1 Transistors, along with other solid state and magnetic devices shall be employed insofar as is practical in order to achieve minimum size, weight, power consumption and a high degree of reliability. Specific approval shall be obtained from the bureau or agency concerned for the use of any electron tubes.

3.2 General. - The equipment shall meet the requirements of Specification MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in Specification MIL-E-16400 for reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

FSC 5805

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3.2.1 Power source. - Equipment shall operate within the requirements specified herein when connected to a source of 50 and 60 cycles per second, ±5 percent, single phase alternating current power at 115 volts, ±10 percent.

3.2.1.1 Fuses. - Fuses with blown-fuse indicators shall be provided in each side of the primary power circuit.

3.2.2 Enclosure. - The equipment shall be housed in three separate dripproof enclosures in accordance with Standard MIL-STD-108 subdivided in the following manner:

(a) Transmit group - which shall contain the complete circuitry of the transmitting station with the exception of the power supply and master oscillator.
(b) Receive group - which shall contain the complete circuitry of the receiving station with the exception of the power supply and master oscillator.
(c) Power supply/master oscillator group - which shall supply the necessary power to operate either two transmit groups, two receive groups or one transmit and one receive group and in addition shall furnish the necessary drive frequency to the transmit and receive groups.

3.2.2.1 Groups of the equipment shall be constructed as frames holding vertical pull-out sections. The pull-out sections shall be mounted in such a manner that all external connections shall be made at the rear and the sections shall be removable from the front of the equipment.

3.2.2.2 Each section shall be operable in the extended position without the use of special cables or plugs.

3.2.2.3 The equipment shall be fully operable with any number of sections in the extended position and the act of pulling out the sections shall cause no interference with the operation of the equipment.

3.2.2.4 The equipment shall be designed to provide for mounting the groups in a standard 19 inch relay rack conforming to Standard MIL-STD-189.

3.2.3 Accessibility. - The equipment shall be so designed that all installation, maintenance and operating procedures may be performed from the front of the cabinet and the cabinet can be mounted with its rear surface and side surfaces against walls or other equipment.

3.2.4 Temperature. - The equipment shall meet the temperature requirements of class 4 of Specification MIL-E-16400.

3.2.5 Shock, vibration and inclination. - The equipment shall meet the shock, vibration, and inclination requirements of Specification MIL-E-16400 for shipboard equipment.

3.2.6 Finish. - External finish of the equipment shall be as specified in Specification MIL-E-16400.

3.2.7 Tropicalization. - Tropicalization will not be required.

3.2.8 Ventilation. - A ventilating system shall be included in the cabinet only if required to meet the temperature requirements.

3.2.9 Test facilities. - Suitable test points and test facilities shall be built into the equipment to enable ease of operation and maintenance without the use of specialized test equipment other than specified herein.

3.2.9.1 A converter neon light test device for visual observation of transmitting and receiving code converters shall be furnished.

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3.2.9.2 A reed-type start-stop oscillator frequency calibration meter shall be furnished as an integral part of the power supply/master oscillator group.

3.2.10 Radio interference and undesired radiation. - The equipment shall meet the radio interference and undesired radiation requirements of Specification MIL-E-16400.

3.3 Functional requirements. -

3.3.1 It shall be the function of the transmitting equipment to accept neutral start-stop signals from 2, 3 or 4 individual teletypewriter channels, and to convert and assemble the information in time sequential order for multiplex transmission over a single circuit.

3.3.2 It shall be the function of the receiving equipment to perform complementary functions to the transmitting equipment, to regenerate and redistribute the received multiplex signal in start-stop form to a like number of teletypewriter channels.

3.3.3 It shall be the function of the local transmitting equipment to operate in conjunction with a distant receiving equipment and the local receiving equipment to operate in conjunction with a distant transmitting equipment.

3.3.4 The associated teletypewriter equipment shall be standard in all respects and the associated local circuitry shall be of any length consistent with normal practice.

3.3.5 The multiplex signals shall be completely compatible for 4-channel inter-operation with the AN/FGC-5 at speeds of 60 and 75 words per minute.

3.4 Detail requirements. -

3.4.1 Start-stop input circuits. -

3.4.1.1 The equipment shall provide facilities for handling 2, 3 or 4 independent start-stop teletypewriter circuits. Variations in the channel numbers shall be accomplished by the operation of a single switch.

3.4.1.2 The input circuits to the transmitting equipment shall be neutral start-stop signals of either 20 or 60 milliamperes direct current with battery supplied from an external source, which may be either positive grounded, negative grounded, or ungrounded.

3.4.1.2.1 Input operating current accommodation shall be fixed for either 20 or 60 milliampere operation by means of a wire strapping arrangement.

3.4.1.2.2 Means shall be provided for fine adjustment of the input current levels.

3.4.1.3 The multiplex shall be operable with associated input equipment operating at 60, 75, or 100 words per minute.

3.4.1.3.1 Operating speeds shall be fixed but may be easily changed by an adjustment/or switching arrangement, or a combination of both.

3.4.1.3.2 Adjustments required as a result of a change in speed shall be kept to a minimum and shall require no test equipment other than the test facilities built into the equipment.

3.4.1.3.3 All channels of the equipment shall be operated at the same speed.

3.4.1.4 The input circuitry shall be such as to permit operation with input signals containing distortion up to 45 percent, either marking or spacing.

3.4.1.5 The input terminals shall be electrically isolated from chassis and cabinet ground and from all other circuits in the equipment.

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3.4.2 Multiplex output. -

3.4.2.1 The multiplex output signal shall consist of a single channel of equal length pulses at a rate dependent upon the number of operating channels and the speed of those channels.

3.4.2.1.1 The multiplex output signal shall contain six intelligence pulses for each channel being transmitted.

3.4.2.1.2 The pulses representing any particular channel shall be transmitted in sequence. (The channels designated "A", "B", "C", and "D" inclusive shall therefore be transmitted in sequential order of A1, A2, A3, A4, A5, A6, B1, B2, B3, B4, B5, B6, C1, C2, C3, C4, C5, C6, D1, D2, D3, D4, D5, D6, A1, A2, A3, and so forth.)

3.4.2.1.3 No separate synchronizing or timing pulses shall be transmitted.

3.4.2.1.4 Provision shall be made for the transmission of inverted signal pulses in a pre-arranged pattern.

3.4.2.2 The output signal shall consist of a keyed neutral direct current signal.

3.4.2.2.1 The transmitter output circuit shall be capable of providing 130 volts into a 1200-ohm load with a maximum internal impedance of 800 ohms.

3.4.2.2.1.1 The internal battery shall be ungrounded, either polarity of which may be grounded externally.

3.4.2.2.2 An additional output between 30 and 45 volts for mark and zero plus or minus 1 volt for space shall be provided across an external impedance of a minimum of 10,000 ohms.

3.4.2.2.3 Means shall be provided for fine adjustment of the output current level (see 3.4.2.2.1) and output voltage level (see 3.4.2.2.2).

3.4.3 Multiplex input. -

3.4.3.1 The receiving equipment multiplex input shall accept neutral keyed 20 or 60 milliampere direct current signals with battery supplied from an external source, which may be either positive grounded, negative grounded, or ungrounded.

3.4.3.2 The input terminals shall be electrically isolated from chassis and cabinet ground and from all other circuits in the equipment.

3.4.3.3 Means shall be provided for fine adjustment of the input current level.

3.4.3.4 The multiplex input circuitry shall be such as to accept the multiplex output signal of 3.4.2.

3.4.4 Start-stop output circuits. -

3.4.4.1 The receiving equipment start-stop output shall be keyed by means of a relay or equivalent circuitry with line battery supplied from an external source.

3.4.4.2 Output signal distortion shall be kept to a minimum and shall not exceed 5 percent.

3.4.4.3 The start-stop operating speed shall be as follows for the three speeds of operation:

Operating speed	Output speed
60 words per minute	375 operations per minute
75 words per minute	468.75 operations per minute
100 words per minute	611 19/46 operations per minute

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3.4.5 Auxiliary signals. -

3.4.5.1 Auxiliary output signals. -

3.4.5.1.1 The following output signals shall be provided by the transmit group for use with auxiliary equipment.

3.4.5.1.1.1 A signal consisting of positive going pulses having minimum amplitudes of 25 volts, originating in an impedance of not greater than 10,000 ohms. The time duration of each pulse shall not be less than 150 microseconds. The pulses shall have a repetition rate as follows and shall be made available on pin D of the connector specified in 3.4.5.3.1.

Operating speed	Repetition rate
60 words per minute	450 p.p.s.
75 words per minute	562.5 p.p.s.
100 words per minute	732 p.p.s.

3.4.5.1.1.2 A signal corresponding to the normal multiplex output signal but inverted. The pulse amplitudes of this signal shall be developed across a load of 220,000 ohms and shall be 0 volts +2 volts for the space signal, and a minimum of +37 volts and a maximum of +55 volts for the mark signal. This signal shall be made available on pin E of the connector specified in 3.4.5.3.1.

3.4.5.1.1.3 A signal consisting of negative going pulses producing a minimum amplitude of 15 volts across a 1 megohm load. The minimum time duration of each pulse shall be 7 microseconds. The repetition rate shall be equal to the multiplex signal baud and the pulses shall occur within 30 microseconds of the start of each code element of the inverted multiplex signal output specified in 3.4.5.1.1.2. This signal shall appear at pin A of the connector specified in 3.4.5.3.1.

3.4.5.1.1.4 A signal consisting of a positive pulse having an amplitude of between +32.5 volts and +42 volts originating in an impedance not greater than 10,000 ohms. The time duration of the pulse shall be not less than 150 microseconds. The pulse shall occur within 30 microseconds after the start of the first code element in "B" channel appearing in the inverted multiplex output signal specified in 3.4.5.1.1.2. This signal shall appear at pin B of the connector specified in 3.4.5.3.1. The rise time of this pulse shall not exceed 1 millisecond.

3.4.5.1.2 The following output signals shall be provided by the receive group for use with auxiliary equipment.

3.4.5.1.2.1 A signal identical to the one specified in 3.4.5.1.1.1 and which shall appear on pin B of the connector specified in 3.4.5.3.2.

3.4.5.1.2.2 A signal identical to the one specified in 3.4.5.1.1.2 and which shall appear on pin F of the connector specified in 3.4.5.3.2.

3.4.5.1.2.3 A signal consisting of a positive pulse having an amplitude of between +32.5 volts and +42 volts originating in an impedance not greater than 10,000 ohms. The time duration of the pulse shall not be less than 150 microseconds. This pulse shall occur within the first half of the sixth code element of "A" channel appearing in the inverted multiplex input signal received from the auxiliary equipment referred to in 3.4.5.2.2, and shall occur only when the sixth code element "A" channel in the multiplex signal being received at the input to the multiplex equipment is a "Mark". This signal shall appear on pin J of the connector specified in 3.4.5.3.2. The rise time of this pulse shall not exceed 1 millisecond.

3.4.5.1.2.4 A signal consisting of negative going pulses having an amplitude of not less than 15 volts, originating in an impedance not greater than 10,000 ohms. Each pulse shall have a time duration of less than 5 percent of multiplex code element width. The pulses shall occur in coincidence with the nominal center of the code elements appearing in the inverted multiplex output signal specified in 3.4.5.1.1.2. This signal shall appear on pin G of the connector specified in 3.4.5.3.2.

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3.4.5.2 Auxiliary input circuits. -

3.4.5.2.1 The transmit group shall provide a d.c. coupled input circuit capable of accepting an inverted multiplex input signal originating in the auxiliary equipment. This signal shall have the following amplitudes:

Space - 0 ± 2 volts
Mark - +55 +20 -10 volts

The input impedance of this circuit shall be 50,000 ohms or greater. This signal shall be accepted on pin F of the connector specified in 3.4.5.3.1.

3.4.5.2.2 The receive group shall provide a d.c. coupled input circuit capable of accepting an inverted multiplex input signal originating in the auxiliary equipment. This signal shall have the following amplitudes:

Space - 0 ± 2 volts
Mark - +90 +20 -10 volts

The input impedance of this circuit shall be 50,000 ohms or greater. The input signal will be delayed 1/2 of a code element from the inverted multiplex output signal specified in 3.4.5.1.1.2. This signal shall be accepted on pin E of the connector specified in 3.4.5.3.2.

3.4.5.3 Signal terminations. - Cable connectors shall be provided for the signals of 3.4.5.1 and 3.4.5.2 separate from the cable connectors provided for normal input, output, power and inter-connection cables. A dummy connector shall be provided to terminate the equipment when the auxiliary equipment is not used.

3.4.5.3.1 The cable connector for the transmitting group shall be a male AN connector, 10 pin (AN3102A-18-1P).

3.4.5.3.1.1 Pin H of this connector shall be connected to the equipment "Ground".

3.4.5.3.2 The cable connector for the receiving group shall be a female connector, 10 pin (AN3102A-18-1S).

3.4.5.3.2.1 Pin I of this connector shall be connected to equipment "Ground".

3.4.6 Additional requirements. -

3.4.6.1 Stability. - The maximum frequency drift of the crystal oscillator shall be one part in one million (10^6) per day.

3.4.6.1.1 Provision shall be included for driving the multiplex from an external frequency standard which has been synthesized externally to derive the normal multiplex driving frequency.

3.4.6.2 Phasing. - Phasing of the equipment shall be accomplished by either of the following methods. The selection of the method shall be accomplished by the operation of a switch:

(a) By the manual operation of a phasing button, and by observing copy received on a receiving teletypewriter, it shall be possible to achieve an in-phase condition.
(b) By interrupting traffic on a designated multiplex channel, it shall be possible, by depressing a button, to automatically achieve an in-phase condition. When an "in-phase" condition has been established, phasing action will automatically cease.

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3.4.6.3 Switching shall be included for operation of the equipment for local, or closed loop operation for circuit lineup purposes.

3.4.6.4 Facilities shall be provided for circuit lineup purposes to transmit the following signals:

(a) Alternate marking and spacing impulses.
(b) Steady mark signal.
(c) Steady space signal.

3.4.6.5 Monitoring facilities. - Circuit monitoring facilities shall be provided for normal operation of the equipment.

3.4.6.5.1 The following circuits shall be monitored:

(a) Input start-stop line current in each channel.
(b) Output multiplex signal line current.
(c) Input multiplex signal line current.
(d) Output start-stop line current in each channel.
(e) Synchronizer operation.
(f) Signal distributor, transmitting and receiving.
(g) All voltage supplies.

3.4.6.6 Synchronization. - Facilities shall be provided at the receiving equipment for maintaining proper synchronism, or phase relationship, between the distant transmitting and local receiving equipment.

3.4.6.6.1 The synchronizer shall function by comparing the time-phase relationship of the incoming signal transitions with the receiving multiplex drive.

3.4.6.6.2 The synchronizer shall have the characteristics specified hereinafter.

3.4.6.6.2.1 A signal integration or averaging circuit shall be included in order to minimize the effects of sudden changes in phase relationship between the transmitted signal and the receiving equipment drive.

3.4.6.6.2.2 Synchronization shall be accomplished without changing the natural oscillating frequency of the frequency standard, thus making the use of a single crystal to drive two or more transmitting and receiving terminals possible.

3.4.6.7 Provision shall be included in the design for completely independent operation of the sending and receiving equipment when each equipment is provided with its own power supply/master oscillator group.

3.5 Workmanship. - Workmanship shall be in accordance with Specification MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

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4.2 Preproduction inspection. - Preproduction inspection shall be performed by the contractor and shall be in accordance with Specification MIL-E-16400 with the following exceptions:

(a) Accelerated life test - In lieu of the accelerated life test, a 15 day operating test shall be run at room temperatures. The allowable failure rate shall be the same as for the accelerated life test.
(b) Shock, vibration and inclination tests and salt spray test may be performed by the Government at a Government laboratory if required by the bureau or agency concerned.

4.3 Acceptance inspection. - Acceptance inspection shall be in accordance with Specification MIL-E-16400 except as specified herein.

4.3.1 Production control inspection. -

(a) Enclosure test shall not be required.
(b) Radio interference and undesired radiation test shall be performed on one production equipment selected by the Government inspector at random from the total number of production equipments offered for delivery and tests shall be repeated if changes in design affect radiation characteristics. These tests may be repeated at the discretion of the Government inspector or the bureau or agency concerned.

4.3.1.1 Sampling for production control inspection. - Sample equipments shall be selected at random by the Government inspector as shown in table I for the tests as specified in 4.3.

Table I - Sampling for production control inspection.

Cumulative number of equipments produced	Cumulative number of sample equipments to be tested	Cumulative number of samples for heat test only
1 to 3	2	1
4 to 6	3	1
7 to 10	5	1
11 to 15	6	2
16 to 25	8	2
26 to 40	10	2
41 to 65	12	3
66 to 95	14	3
96 to 145	17	3
146 to 200	20	4
201 to 290	24	4
291 to 440	30	4

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking (see 6.1). -

5.1.1 The equipment shall be preserved and packaged by level A or C; packed by level A, B or C and marked by levels A and B or C, as specified in the contract or order in accordance with Specification MIL-P-17555.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Levels of shipment required (see 5.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships

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